CONVERSION

Convert a hex color to RGB

Example Function:

```
hex_to_rgb() {
    # Usage: hex_to_rgb "#FFFFFF"
    #        hex_to_rgb "000000"
    : "${1/\#}"
    ((r=16#${_:0:2},g=16#${_:2:2},b=16#${_:4:2}))
    printf '%s\n' "$r $g $b"
}
```

Example Usage:

```
$ hex_to_rgb "#FFFFFF"
255 255 255
```

Convert an RGB color to hex

Example Function:

```
rgb_to_hex() {
    # Usage: rgb_to_hex "r" "g" "b"
    printf '#%02x%02x%02x\n' "$1" "$2" "$3"
}
```

Example Usage:

```
$ rgb_to_hex "255" "255" "255"
#FFFFFF
```

CODE GOLF

Shorter `for` loop syntax

```
# Tiny C Style.
for((;i++<10;)){ echo "$i";}

# Undocumented method.
for i in {1..10};{ echo "$i";}

# Expansion.
for i in {1..10}; do echo "$i"; done

# C Style.
for((i=0;i<=10;i++)); do echo "$i"; done
```

Shorter infinite loops

```
# Normal method
while :; do echo hi; done

# Shorter
for((;;)){ echo hi;}
```

Shorter function declaration

```
# Normal method
f(){ echo hi;}

# Using a subshell
f()(echo hi)

# Using arithmetic
# This can be used to assign integer values.
# Example: f a=1
#          f a++
f()(($1))

# Using tests, loops etc.
# NOTE: ‘while’, ‘until’, ‘case’, ‘(())’, ‘[[]]’ can also be used.
f()if true; then echo "$1"; fi
f()for i in "$@"; do echo "$i"; done
```

Shorter `if` syntax

```
# One line
# Note: The 3rd statement may run when the 1st is true
[[ $var == hello ]] && echo hi || echo bye
[[ $var == hello ]] && { echo hi; echo there; } || echo bye

# Multi line (no else, single statement)
# Note: The exit status may not be the same as with an if statement
[[ $var == hello ]] &&
    echo hi

# Multi line (no else)
[[ $var == hello ]] && {
    echo hi
    # ...
}
```

Simpler case statement to set variable

The `:` built-in can be used to avoid repeating `variable=` in a case statement. The `$_` variable stores the last argument of the last command. `:` always succeeds so it can be used to store the variable value.

```
# Modified snippet from Neofetch.
case "$OSTYPE" in
    "darwin"*)
        : "MacOS"
    ;;

    "linux"*)
        : "Linux"
    ;;

    *"bsd"* | "dragonfly" | "bitrig")
        : "BSD"
    ;;

    "cygwin" | "msys" | "win32")
        : "Windows"
    ;;

    *)
        printf '%s\n' "Unknown OS detected, aborting..." >&2
        exit 1
    ;;
esac

# Finally, set the variable.
os="$_"
```